DevOps in the Age of AI

Human Powered Evolution

Kellyn Pot'Vin-Gorman
Engineer and Advocate | AI Advisor | Multiplatform Database
Redgate

Kellyn Gorman

- *30 years in tech industry*
- *Engineer and Advocate covering multiplatform and AI at Redgate*
- *Formerly with Microsoft and Oracle.*
- *Runs PDXWIT in Portland, Oregon*
- *AI Advisor as outside role*
- *Presenter, author, blogger, podcaster and tech contributor as DBA Kevlar*

Over 70% of companies now prioritize AI initiatives as a core corporate strategy.

https://thecuberesearch.com/the-state-of-devops-and-ais-growing-influence/

30 Years of BASH. Then Everyone Wanted PowerShell.

My Background

Oracle DBA. Linux native. BASH and Python were my first languages and BASH my most fluent one.

The New Reality

At Redgate, my audience was Windows-first. DBAs, DevOps engineers, platform teams. All living in PowerShell. I needed to speak their language.

The Problem

Traditional tutorials assume you start from zero. I didn’t need to start from zero, but I needed a translator.

```
# BASH (what I knew)
for f in *.sql; do
  psql -f "$f"
done

# PowerShell (what they needed)
Get-ChildItem *.sql | ForEach-Object {
  psql -f $_.FullName
}
```

Same logic. Different dialect.

AI as a Translator, Not a Teacher

01

The Prompt

“I know how to do this in BASH. Show me the PowerShell equivalent and explain why it works differently.”

Starting from what I already understood made every answer immediately useful.

02

The Method

Paste a BASH command I understood. Ask AI to translate it to PowerShell. Ask it to explain the structural difference, not just give me the answer.

Understanding the why meant I wasn’t copy-pasting. I was actually learning.

03

The Result

Weeks, not months. I could write and debug real PowerShell scripts for Windows audiences and fast enough to be useful, confident enough to teach it.

My repos are no longer BASH, Python and SQL.

What AI Actually Accelerated

Weeks

to functional fluency, not months

Zero-to-productive in a scripting language I had actively avoided for years. Yes, I worked at Microsoft for over 5 years and had continued scripting in other languages.

Context-aware learning is faster learning

Because I gave AI my existing knowledge as the anchor, every answer landed in context I already had. No ramp-up. No wasted explanation of things I already knew.

This is not just my story

Every engineer in this room has a skill gap they've been putting off. A cloud provider they haven't touched. A language they avoid. A framework that intimidates them.

AI was available when I was, on my schedule. It didn't question my questions or challenge my method of learning.

AI doesn't replace what you know. It builds on it.

The 2024 Stack Overflow developer survey found that over 80% of developers cited improved productivity as the top benefit of AI tools, with about 62% saying AI sped up learning and 59% citing greater efficiency.

https://www.statista.com/statistics/1440348/ai-benefits-in-development-workflow-globally/

If AI can teach a Linux DBA to love PowerShell...

imagine what it can do for Developers.

AI as a learning accelerator is just the beginning. Let's talk about what it means for DevOps at scale.

AI is changing everything, including DevOps.

The question isn't whether DevOps survives, but how it might thrive.

3×

Faster IaC provisioning with AI assistance

40%

Reduction in manual operational toil

∞

Human judgment still required at the edge

How Much is it Changing?

- 4.2 million developers as of early 2025, AI-authored code accounted for 27% of all production code
- Up 5-11% each quarter.
- Daily AI users have nearly a third of their merged, production-ready code written by AI.

https://shiftmag.dev/this-cto-says-93-of-developers-use-ai-but-productivity-is-still-10-8013/

WHAT AI BRINGS TO DEVOPS

Generative IaC

AI drafts Terraform, CloudFormation & Bicep scripts, which engineers review, validate, and optimize.

Policy & Drift Detection

Automated compliance enforcement flags misconfigurations before they reach production.

Intelligent Anomaly Detection

AIOps surfaces issues in real-time, correlating signals across multi-environment telemetry.

Faros AI's telemetry across 10,000+ developers found a 21% increase in tasks completed and a 98% increase in pull requests merged.

https://www.faros.ai/blog/key-takeaways-from-the-dora-report-2025

WHAT AI BRINGS TO DEVOPS

Auto-Remediation

Intelligent runbooks resolve known failure patterns without waking up the on-call engineer.

Accelerated CI/CD

AI suggests pipeline optimizations, identifies bottlenecks, and predicts build failures early.

Predictive Capacity Planning

ML-driven forecasting replaces reactive scaling with proactive infrastructure decisions.

- Developers' self-reports showed a savings of around 4 hours per week, including onboarding time.
- These were measured by time to the 10th pull request and was cut roughly in half since Q1 2024.

https://shiftmag.dev/this-cto-says-93-of-developers-use-ai-but-productivity-is-still-10-8013/

WHERE AI HITS THE WALL

AI lacks the contextual awareness to orchestrate complex, multi-environment systems.

Advanced Reasoning

Multi-variable trade-offs across cost, risk, compliance, and team capacity require human synthesis and not pattern matching.

Complex Orchestration

Cross-team dependencies, legacy integrations, and edge-case failures require institutional knowledge and human judgment.

The gap between what AI can generate and what systems actually need is exactly where DevOps professionals live.

Organizations with fragmented tooling or unclear processes can experience less than satisfactory ROI. AI accelerating technical debt accumulation and introducing instability into already fragile systems.

https://www.infoq.com/news/2026/03/ai-dora-report/

WHERE AI HITS THE WALL

AI lacks the contextual awareness to orchestrate complex, multi-environment systems.

Edge Case Failures

Novel failure modes, cascading incidents, and unknowns fall outside training data and engineers must drive recovery.

Architectural Resilience

Defining guardrails, validating AI outputs, and ensuring system-wide coherence remain deeply human responsibilities.

The gap between what AI can generate and what systems actually need is exactly where DevOps professionals live.

A 2025 randomized controlled trial by METR on experienced open-source developers found the counterintuitive result that developers using AI tools took 19% longer than those without.

This suggests, for highly experienced engineers working in deeply familiar codebases, AI can actually create friction rather than speed.

https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/

THE DEVOPS PROFESSIONAL IN AN AI WORLD

From engineer to orchestrator of human-AI collaboration.

PROMPT

Engineer

Define intelligent prompts that drive accurate IaC generation, pipeline configs, and runbook automation.

REVIEW

Validator

Audit AI-generated outputs for security gaps, architectural flaws, and policy violations before deployment.

DECIDE

Strategist

Make high-stakes architectural and trade-off decisions that require business context and system-wide judgment.

AI magnifies impact. It doesn't diminish it.

A PRACTICAL FRAMEWORK FOR AI-AUGMENTED DEVOPS

01

Define Guardrails First

Before deploying AI tooling, establish policy-as-code, approval gates, and validation criteria. Install tools to remove the temptation of shadow AI.

02

Prompt Engineering as a Discipline

Invest in training engineers to craft precise, context-rich prompts for IaC and pipeline generation.

03

Human Review is Non-Negotiable

All AI-generated configurations require engineer sign-off before production deployment.

04

Build Feedback Loops

Capture AI errors and near-misses. Feed corrections back into your tooling and runbooks.

Why?

Analysis of 153 million changed lines of code found that **"code churn“** or code discarded within two weeks of being written, which AI doubles, and copy/pasted code is increasing faster than updated, deleted, or moved code.

This characterized AI-generated code as similar in composition to a short term developer who doesn't thoughtfully integrate their work into the broader project.

https://devops.com/ai-in-software-development-productivity-at-the-cost-of-code-quality-2/

A PRACTICAL FRAMEWORK FOR AI-AUGMENTED DEVOPS

Key Principles

- Trust but verify… always
- AI accelerates; humans steer
- Governance before automation
- Prompt quality = output quality
- Technical debt and data quality is essential

IAC STRATEGIES: PROVISION · SECURE · TEST AT SCALE

PROVISION

AI-Accelerated Infrastructure

- Generative Terraform / CloudFormation via Copilot, Pulumi AI
- AI suggests right-sizing based on workload profiles
- Drift detection auto-flags config divergence
- Natural language → IaC template in seconds

The IaC lifecycle is the first place AI creates leverage — and the first place human review must be non-negotiable.

IAC STRATEGIES: PROVISION · SECURE · TEST AT SCALE

SECURE

Policy as Code, Enforced by AI

- Checkov / Bridgecrew scan generated scripts pre-deploy
- AI flags IAM over-privilieged and open security groups
- Compliance guardrails (SOC2, HIPAA, CIS) embedded in pipelines
- Secrets detection prevents credentials leaking into IaC

The IaC lifecycle is the first place AI creates leverage — and the first place human review must be non-negotiable.

IAC STRATEGIES: PROVISION · SECURE · TEST AT SCALE

TEST

Validation Before It Ever Ships

- AI-generated integration tests from IaC spec
- Canary + blue-green auto-routing with anomaly rollback
- Contract testing between services via AI inference
- Chaos engineering suggestions based on dependency maps

The IaC lifecycle is the first place AI creates leverage — and the first place human review must be non-negotiable.

BLUEPRINTS FOR HUMAN-AI SYMBIOSIS

Balancing AI speed with DevOps expertise — the operating model that scales.

WORKFLOW DESIGN

The Loop Model

AI generates → Human validates → System learns

Establish a closed feedback loop: AI drafts IaC or pipeline logic, engineers review and annotate errors, and those corrections retrain or refine the AI tooling. Speed improves every cycle without sacrificing control.

BLUEPRINTS FOR HUMAN-AI SYMBIOSIS

Balancing AI speed with DevOps expertise — the operating model that scales.

GOVERNANCE

The Co-Pilot Contract

Define what AI owns vs. what humans own

Document explicit ownership boundaries. AI owns: first draft generation, anomaly flagging, pattern-matching. Humans own: architectural decisions, compliance sign-off, incident command, and any output touching production.

BLUEPRINTS FOR HUMAN-AI SYMBIOSIS

Balancing AI speed with DevOps expertise — the operating model that scales.

OPERATIONS

The Escalation Ladder

When AI uncertainty triggers human intervention

Define confidence thresholds. Below 90% AI confidence → auto-flag for review. Novel environment patterns → mandatory human review. Multi-region or regulated workloads → human approval gate before any automated action.

Symbiosis isn't accidental — it's architected. Define the model before you deploy the tool.

BLUEPRINTS FOR HUMAN-AI SYMBIOSIS

Balancing AI speed with DevOps expertise — the operating model that scales.

TEAM DESIGN

The Skills Multiplier

AI raises the floor; engineers raise the ceiling

Junior engineers use AI scaffolding to operate at mid-level productivity. Senior engineers shift from writing boilerplate to defining architecture and reviewing AI outputs. The entire team's effective output scales without headcount.

Symbiosis isn't accidental — it's architected. Define the model before you deploy the tool.

CERTIFYING & REFINING AI-GENERATED AUTOMATION

CERTIFICATION PIPELINE

1. **GENERATE**
   AI produces IaC template, pipeline config, or runbook draft
2. **SCAN**
   Automated policy checks: security, cost, compliance, drift
3. **REVIEW**
   Engineer validates intent, architecture fit, and edge cases
4. **STAGE**
   Deploy to non-prod; AI monitors for anomalies vs. baseline
5. **CERTIFY**
   Sign-off gate: human approval + automated test pass required

Certification isn't bureaucracy — it's the trust layer that lets AI automation scale safely.

CERTIFYING & REFINING AI-GENERATED AUTOMATION

Aligning the "what" and "how" of AI automation with organizational goals.

REFINEMENT PRINCIPLES

Alignment Filter

Before certifying any automation, validate against three questions: Does it serve the stated business goal? Does it respect team constraints? Does it leave an audit trail?

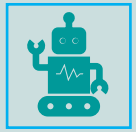
Continuous Calibration

Track AI output quality over time. Score each certified template: accuracy, security pass rate, rework rate. Feed low-scoring outputs back as negative training examples.

Organizational Fitness Test

Every 90 days, review AI-generated automation inventory. Retire stale patterns, update guardrails for new compliance requirements, and re-certify anything that has drifted.

Certification isn't bureaucracy — it's the trust layer that lets AI automation scale safely.



AI is the most powerful tool in your DevOps arsenal.

It will **amplify** you if its done correctly and the environment is prepared.

2025 DORA Report on AI

One analyst called AI **"the mirror principle."** AI is like a mirror that reflects and amplifies your organization's true strengths and challenges.

https://adamferrari.substack.com/p/ai-insights-from-the-2025-dora-report

For teams with:

- solid foundations, AI is a force multiplier.
- broken processes and dysfunctional systems, AI magnifies the chaos.

From this, they discovered…

The greatest returns on AI investment come not from the tools themselves, but from a strategic focus on the quality of internal platforms, the clarity of workflows, and the alignment of teams

https://circleci.com/blog/dora-ai-amplifier/

THE FUTURE IS YOURS TO STEER

Upskill Your Team

Invest in prompt engineering, AI governance, and review workflows as core competencies.

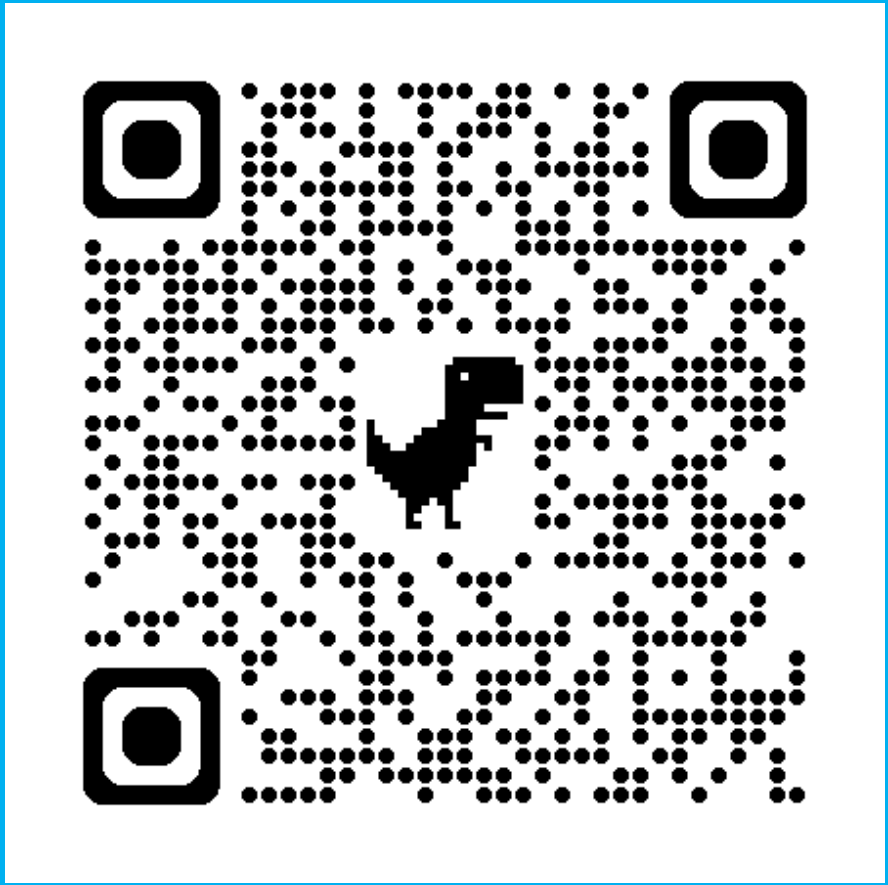
Build Guardrails

Establish policy-as-code and AI validation gates before scaling automation. Use tools to stop shadow AI.

Experiment Boldly

Adopt AI-powered IaC and AIOps tooling in low-risk environments to learn fast, scale smart.

Thank you!



Kellyn Gorman
Redgate Engineer and Advocate
Connect with me:

Jane Street

fleet

Chocolatey

PATCH MY PC

PDQ

SynEdgy